The interpolation law of the uniform doubling condition for curved truncations

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Abstract

In companion papers the critical scaling of the uniform doubling condition of [Shmalo, *Ann. Math. Artif. Intell.* (2024)] for i.i.d. samples was determined to be $\Theta\big((\log N/N)^{2/(2n-j)}\big)$ at every transversal boundary stratum of dimension j, first in the orthogonal model and then for all simplicial (half-space) truncations in general position, the upper bound resting on a witness-box section lemma that diagonalizes the active constraints. Those arguments use the flatness of the truncating surfaces in two places: the level constraint of a half-space is linear, and the slice of a half-space by a coordinate hyperplane is again a half-space. Here we remove both uses for a *fixed* smooth strictly convex truncation family. Let $G = K \cap \bigcap_{i=1}^{k}\{g_i \le 0\}$ with K and the g_i fixed and C^2, the surfaces $\{g_i = 0\}$ and ∂K strictly convex and meeting transversally in general position along a j-dimensional stratum. We prove that the cap of depth h at the stratum has volume $\asymp h^{(2n-j)/2}$ with the same exponent as in the flat case, the constant now depending on the joint second fundamental form of K and the g_i but bounded above and below for fixed C^2 data; and that the slice of G by a level set of one constraint is an admissible $(n-1, k-1)$ curved configuration, with curvature bounds depending on the fixed configuration and uniform along the sliced stratum, closing the slicing induction. Consequently the interpolation law $\Theta\big((\log N/N)^{2/(2n-j)}\big)$ holds for the fixed curved family. The mechanism that survives curvature is the steep/shallow trichotomy at the angular threshold $\sqrt{h}$: a strictly convex constraint contributes a factor h through its linear part when the slab direction is transversal to its normal, and degrades continuously to a factor $\sqrt{h}$ exactly when the slab direction approaches that normal, at which point a lower-dimensional stratum becomes active and the exponent law reports the correct larger value. We isolate honestly what is and is not obtained: for a fixed family the constants are uniform over the stratum and over directions in any fixed compact subcone of the active normal cone, which suffices for the fixed-family interpolation law; a curvature-band-uniform lower/slab constant near the multiply-tangent directions remains open, and we state the residual estimate precisely. We also prove a conditional uniform theorem: if iterated slices keep a uniform curvature floor and all simultaneously projected near-tangent normals have a uniform Gram lower bound, then the slab constants and the interpolation law are uniform over that resolved subfamily. As an immediate consequence, every finite atlas of fixed curved configurations, and every compact atlas whose resolved constants vary continuously and stay positive, has a uniform law with atlas-dependent constants. Resolved geometry is also stable under C^2-small perturbations, so any compact C^2-continuous resolved atlas is uniform by a finite subcover. The same statement is locally uniform on every compact stage of a σ-compact resolved atlas, and becomes globally uniform exactly when the resolved constants have a positive global floor. For one curved truncation, the resolved hypotheses are automatic under quantitative

splaying, and also under an explicit curvature-dominance inequality that allows controlled oblique intersections; in particular, nonobtuse one-cut contact is uniform.

1 Setting and results

We keep the framework of the companion papers [3, 4, 5, 6, 7]. Let $K \subset \mathbb{R}^n$ be a convex body, μ a probability measure with density bounded above and below on K, and $T = \{X_1, \dots, X_N\}$ i.i.d. samples with empirical measure $\bar{\mu}_N$. For a unit vector u, center s, width w and a truncation region G, the truncated slab is $S_w = \{x : |\langle x - s, u\rangle| \le \frac{w}{2}\} \cap G$, and the uniform doubling condition (UDC) with constants $(\kappa, \delta, \sigma, \beta, \ell)$ requires $\bar{\mu}_N(S_{\ell\kappa^{i+1}}) \ge \min\{\delta, (1+\sigma)\bar{\mu}_N(S_{\ell\kappa^i})\}$ for every truncated slab generated by the family and every grid scale $\ell\kappa^i < \beta$. By [4, 5, 6, 7] the critical scaling of ℓ for half-space truncation families in general position is $\Theta\big((\log N/N)^{2/(2n-j)}\big)$, where j is the dimension of the lowest transversal boundary stratum; the lower bound comes from a second-moment count over $\asymp (N/\log N)^{j/(2n-j)}$ congruent caps along the stratum, and the upper bound from a deterministic slab-volume estimate $\mathrm{vol}(G \cap \{|\langle y - x, u\rangle| \le L\}) \gtrsim L^{(2n-j)/2}$ fed into the proof skeleton of [3].

The immediately preceding companion [8] removes the orthogonality and roundness of the orthogonal model, treating a C^2 strictly convex body K cut by half-spaces in general position, and reduces the interpolation law to the flat-truncation case; it leaves open exactly the configurations whose truncating *surfaces* are themselves curved. This note treats that case.

Definition 1.1 (Admissible curved configuration)**.** *Fix $n \ge 2$ and $0 \le k \le n-2$, and put $j := n-1-k$. Let $K \subset \mathbb{R}^n$ be a compact convex body with C^2 boundary whose principal curvatures lie in $[c_K, C_K]$, $0 < c_K \le C_K$, and let $g_1, \dots, g_k \in C^2(\mathbb{R}^n)$ be convex, with*

$$0 < \gamma_0 \le |\nabla g_i| \le \Gamma_0, \qquad c_g\, I \preceq \operatorname{Hess} g_i \preceq C_g\, I \quad \text{on a neighborhood of } G, \tag{1}$$

($0 < c_g \le C_g$), so that each $\{g_i = 0\}$ is a C^2 strictly convex hypersurface. Write $G := K \cap \bigcap_{i=1}^k \{g_i \le 0\}$ and let $n_i := \nabla g_i/|\nabla g_i|$ be the outward unit normal of the i-th surface. We call the data admissible *if G has nonempty interior, contains a ball of a fixed radius $r_0 > 0$, and the surfaces meet* transversally in general position along a j-dimensional stratum

$$E := \partial K \cap \bigcap_{i=1}^{k} \{g_i = 0\},$$

a compact C^2 j-manifold, meaning: at every $p \in E$ the $k+1$ outward unit normals $\nu_K(p), n_1(p), \dots, n_k(p)$ have Gram matrix $M(p)$ with smallest eigenvalue $\ge \gamma > 0$, uniformly in $p \in E$. We call $(c_K, C_K, c_g, C_g, \gamma_0, \Gamma_0, \gamma, r_0)$ the structure constants.

The simplicial model of [6] is the degenerate case $C_g = 0$, $\operatorname{Hess} g_i \equiv 0$ (linear g_i), with $K = B^n$. Our standing example is a ball truncated by two spheres, $K = B^n$ and $g_i(x) = |x - z_i|^2 - \rho_i^2$ with the z_i placed so that the two spheres and ∂K meet transversally along an $(n-3)$-sphere.

For $p \in E$ the maximum of a linear functional over G is governed, to first order, by the active constraints at p. As in [6] the relevant extremal directions are those of the *deepest corner*: u for which the maximizer of $\langle \cdot, u\rangle$ over G is the single point $p \in E$ with *all $k+1$ Karush–Kuhn–Tucker multipliers strictly positive.* These form the relative interior of the normal cone of E at p,

$$\mathcal{U}_p := \operatorname{relint}\Big\{\lambda_K \nu_K(p) + \sum_{i=1}^{k} \lambda_i n_i(p) : \ \lambda_K, \lambda_i \ge 0\Big\} \cap S^{n-1}.$$

Theorem 1.2 (Cap volume at a curved stratum)**.** *Let the data be admissible with structure constants as above, and fix $\theta \in (0, 1]$. There are $h_1 > 0$ and $b_2 > 0$ depending on the fixed configuration data, and $b_1 = b_1(\theta) > 0$ depending in addition on θ, such that for every $p \in E$, every $u \in \mathcal{U}_p$ at sphere-distance at least θ from $\partial\mathcal{U}_p$, and every $0 < h \le h_1\theta^2$,*

$$b_1(\theta)\, h^{\frac{2n-j}{2}} \;\le\; \mu\big(\{x \in G :\ \langle x, u\rangle \ge \tau_{\max}(u) - h\}\big) \;\le\; b_2\, h^{\frac{2n-j}{2}}, \qquad \tau_{\max}(u) := \max_{x\in G}\langle x, u\rangle.$$

The upper constant b_2 is uniform over the whole cone (no θ); the lower constant degrades as $\theta \to 0$ ($b_1(\theta) \gtrsim \theta^{k+j}$), staying positive for each fixed θ. In particular, for u ranging over any compact subcone of $\mathcal{U}_p$ (fixed θ), the cap exponent is exactly $(2n - j)/2$, independent of the curvatures, with constants determined by the fixed configuration data and θ.

Theorem 1.3 (Section lemma and slab bound)**.** *For each fixed configuration satisfying Definition 1.1 there are $a_1 > 0$ and $L_0 > 0$, depending on that configuration, such that for every $x \in G$, every unit u, and every $0 < L \le L_0$,*

$$\mathrm{vol}\Big(G \cap \{y :\ |\langle y - x, u\rangle| \le L\}\Big) \;\ge\; a_1\, L^{\frac{2n-j}{2}}.$$

Corollary 1.4 (Interpolation law, fixed curved family)**.** *Let the data be admissible. Then the critical scaling of the uniform doubling condition for the (single fixed) curved truncation family $\{g_1, \dots, g_k\}$ on K is*

$$\ell_N \;\asymp\; \Big(\frac{\log N}{N}\Big)^{2/(2n-j)}, \qquad j = n - 1 - k,$$

both bounds holding with probability $1-o(1)$, exponent and logarithm exact. The implied constants depend on the fixed configuration data (equivalently, on the structure constants together with the nondegeneracy constants of this family), but not on N.

Section 2 proves Theorem 1.2; Section 3 proves the slicing induction behind Theorem 1.3 and assembles Corollary 1.4; Section 4 states precisely what uniformity is and is not obtained, isolates the residual open estimate, and proves a conditional uniform theorem under resolved multi-tangent geometry.

2 The cap-volume proposition with curved constraints

Throughout this section fix $p \in E$ and $u \in \mathcal{U}_p$ at sphere-distance $\ge \theta$ from $\partial\mathcal{U}_p$. After a fixed rotation and translation we place p at the origin with $u = e_1$ pointing in the direction of increasing $\tau = \langle \cdot, u\rangle$; thus $\tau_{\max}(u) = 0$ and the cap of depth h is

$$C(h) = \{x \in G :\ x_1 \ge -h\}.$$

Write $x = (x_1, \xi)$ with $\xi \in \mathbb{R}^{n-1}$ the slab coordinates (the orthogonal complement of u).

2.1 The local floor

At $p = 0$ the $k + 1$ active surfaces are ∂K and $\{g_i = 0\}$, $i = 1, \dots, k$. Each is a C^2 graph over its tangent plane at 0; since all $k + 1$ outward normals lie in the open half-space $\{x_1 > 0\}$-side (because $u = e_1$ is a strictly positive combination of them), each surface is, near 0, a graph $x_1 = \psi_\bullet(\xi)$ over the slab, with

$$\psi_K(\xi) = -\langle a_K, \xi\rangle - \tfrac{1}{2}\xi^\top Q_K\, \xi + O(|\xi|^3), \qquad \psi_i(\xi) = -\langle a_i, \xi\rangle - \tfrac{1}{2}\xi^\top Q_i\, \xi + O(|\xi|^3), \qquad (2)$$

where $a_\bullet \in \mathbb{R}^{n-1}$ is the slab-gradient of the surface at 0 and $Q_\bullet \succeq 0$ is its second fundamental form in the slab coordinates (positive semidefinite by convexity of K and of the g_i; positive definite on the j stratum directions for K, see below). The $O(|\xi|^3)$ remainders are uniform on E for fixed C^2 data. The region G near 0 is

$$C(h) = \Big\{(x_1, \xi) : \ -h \le x_1 \le \min\big(\psi_K(\xi),\, \psi_1(\xi), \dots, \psi_k(\xi)\big)\Big\}. \tag{3}$$

The *floor* is the function $\Phi(\xi) := \min_\bullet \psi_\bullet(\xi)$, and the cap is the region between the height $-h$ and the floor wherever the floor exceeds $-h$.

The linear data $a_\bullet$ encode the angle between u and the surface normals. Indeed if ν is the outward unit normal of a surface $\{x_1 = \psi(\xi)\}$ at 0, then writing $\nu = (\nu_1, \nu')$ with $\nu_1 = \langle \nu, u\rangle$ one has $a = -\nu'/\nu_1$, hence

$$|a_\bullet| = \frac{|\nu'_\bullet|}{\langle \nu_\bullet, u\rangle} = \frac{\sin\angle(\nu_\bullet, u)}{\cos\angle(\nu_\bullet, u)} = \tan\angle(\nu_\bullet, u), \qquad \beta_\bullet := \frac{1}{\sqrt{1+|a_\bullet|^2}} = \langle \nu_\bullet, u\rangle. \tag{4}$$

The quantity $\beta_\bullet = \langle \nu_\bullet, u\rangle \in (0,1]$ is the cosine of the angle between u and the $\bullet$-normal, and $m_\bullet := |a_\bullet| = \tan\angle(\nu_\bullet, u) = \sqrt{1-\beta_\bullet^2}/\beta_\bullet$ is the slope of the surface graph in the slab coordinates. The slope governs the pinning: along the steepest slab direction $\hat{a}_\bullet := a_\bullet/|a_\bullet|$ the floor $\psi_\bullet$ decreases at unit rate $m_\bullet$, so the constraint $x_1 \le \psi_\bullet$ at height $x_1 \ge -h$ confines that coordinate to a window of length $\asymp h/m_\bullet$. A *steep* surface (slope $m_\bullet$ bounded, u transversal to the normal) has a window of length $\asymp h$; a *shallow* one ($m_\bullet$ large, u near the surface's tangent plane, $\beta_\bullet \to 0$) has a shorter window, but the surface curvature then caps the relevant extent at $\sqrt{h}$. This is the curved analogue of the u_i-window $h/(8ku_i)$ of [6], where $u_i = \langle u, e_i\rangle$ is exactly the cosine β_i for the orthogonal model. We record the two structural facts.

Lemma 2.1 (Curvature is positive definite on the stratum)**.** *Let $V \subseteq \mathbb{R}^{n-1}$ be the slab-image of T_pE, the tangent space of the stratum,* $\dim V = j$*. Then there is a structure constant $c'_K \asymp c_K > 0$ such that for every $p \in E$ and every admissible u,*

$$Q_K\big|_V \ \succeq \ c'_K\, I_V,$$

so the floor is strictly convex along the stratum directions. Consequently the joint tangential form $Q_K\big|_V + \sum_i Q_i\big|_V \succeq c'_K I_V$ as well, since each $Q_i \succeq 0$ by convexity of g_i.

Proof. ∂K has second fundamental form $\mathrm{II}_K \succeq c_K I$ on its tangent plane $T_p\partial K$. Writing ∂K near p as a graph $x_1 = \psi_K(\xi)$ over the slab $u^\perp$, tilted from $T_p\partial K$ by the angle $\alpha = \angle(\nu_K, u)$, the graph Hessian is $Q_K = \mathrm{II}_K/\cos^3\alpha$ in the tilt direction and $\mathrm{II}_K/\cos\alpha$ transversally, hence $Q_K \succeq \mathrm{II}_K \succeq c_K I$ on $u^\perp$ (tilting only increases the graph curvature). Restricting to V, the slab projection $T_pE \to V$ is a linear isomorphism with singular values bounded above by 1, so $Q_K|_V \succeq c'_K I_V$ with a structure constant $c'_K \asymp c_K$ that does *not* degrade as u approaches a facet (the tilt makes Q_K larger, not smaller). The last sentence is immediate from $Q_i \succeq 0$. □

Lemma 2.2 (Transversal lower bound on the corner angle)**.** *There is a structure constant $\beta_* > 0$ (depending on γ and on n, k) such that for every $p \in E$ and every $u \in \mathcal{U}_p$ at sphere-distance $\ge \theta$ from $\partial\mathcal{U}_p$,*

$$\beta_\bullet = \langle \nu_\bullet, u\rangle \ \ge \ \beta_*\,\theta \qquad \textit{for each active normal } \nu_\bullet \in \{\nu_K, n_1, \dots, n_k\}.$$

Proof. Write $u = \sum_\bullet \lambda_\bullet \nu_\bullet$ with $\lambda_\bullet \geq 0$ and, since $|u| = 1$, $\lambda^\top M \lambda = 1$, where $M \succeq \gamma I$ is the Gram matrix. Hence $|\lambda| \leq \gamma^{-1/2}$, and $\langle \nu_\bullet, u \rangle = (M\lambda)_\bullet$. If u lies at sphere-distance $\geq \theta$ from $\partial \mathcal{U}_p$, then u is not within θ of any facet $\{\lambda_\bullet = 0\}$ of the cone; equivalently each $\lambda_\bullet$ is bounded below by a multiple of θ (the facet distance is comparable to the smallest multiplier, with comparison constants fixed by $M \in [\gamma, \Gamma_0^2]$). Thus $(M\lambda)_\bullet \geq \gamma \lambda_\bullet \geq \beta_* \theta$. Consequently the slope is bounded, $m_\bullet = \sqrt{1 - \beta_\bullet^2}/\beta_\bullet \leq 1/\beta_\bullet \leq 1/(\beta_* \theta)$. □

2.2 The witness box

We now run the witness-box construction of [6] with the linear constraint $x_i \leq 0$ replaced by the curved floor (2), and the orthogonality $\langle w, \tilde{u} \rangle = 0$ replaced by working directly in the slab coordinates ξ, in which the height x_1 is the free fibre direction. Split the active surfaces by the slope threshold at the angular scale $\sqrt{h}$: with $C_* := C_g + C_K$ bounding all floor curvatures,

$$\mathcal{A} := \{\bullet : \ m_\bullet \leq C_*^{-1/2} h^{-1/2}\}, \qquad \mathcal{Z} := \{\bullet : \ m_\bullet > C_*^{-1/2} h^{-1/2}\}. \tag{5}$$

We call $\mathcal{A}$ the *steep* surfaces (bounded slope: u transversal to the normal, $\beta_\bullet$ not too small) and $\mathcal{Z}$ the *shallow* (slope $\gtrsim h^{-1/2}$: u near the surface's tangent plane, $\beta_\bullet \lesssim \sqrt{h}$). By Lemma 2.2, $m_\bullet \leq 1/(\beta_* \theta)$ for u at facet-distance $\geq \theta$, so every active surface is steep, $\mathcal{Z} = \varnothing$, once $1/(\beta_* \theta) \leq C_*^{-1/2} h^{-1/2}$, i.e. once $h \leq \beta_*^2 \theta^2 / C_* =: h_1 \theta^2$. This is the depth range asserted in Theorem 1.2; we carry out the argument there. The role of the trichotomy across the transition $\mathcal{Z} \neq \varnothing$ is deferred to Section 4.

Fix $h \leq h_1 \theta^2$, so every $m_\bullet \leq C_*^{-1/2} h^{-1/2}$, equivalently the pinned window $h/m_\bullet \geq \sqrt{h/C_*}\cdot$(a constant) and the linear part of each $\psi_\bullet$ dominates its quadratic part at scale h. Let V be the slab-image of $T_p E$ as in Lemma 2.1, and let $W := V^\perp \subseteq \mathbb{R}^{n-1}$, $\dim W = k$. The slab-projected constraint normals $P_u n_1, \ldots, P_u n_k$ span W; by transversality their Gram is bounded below by a structure constant, so there is a *biorthogonal (dual) frame* $f_1, \ldots, f_k \in W$,

$$\langle f_i, \, P_u n_{i'} \rangle = \delta_{ii'}, \qquad |f_i| \leq F_0 \text{ (a structure constant)}, \tag{6}$$

bounded because the Gram is bounded away from singular. This frame plays the role of the orthogonality $\langle w, \tilde{u} \rangle = 0$ of [6]: it provides coordinates $s_i := \langle \xi, P_u n_i \rangle$ for $\xi \in \mathbb{R}^{n-1}$ in which the i-th constraint depends, to leading order, only on s_i, namely $\psi_i(\xi) = -s_i - \frac{1}{2} \xi^\top Q_i \xi + O(|\xi|^3)$ (the linear part of ψ_i is $-\langle a_i, \xi \rangle$ with $a_i = -P_u n_i / \beta_i$, so $-\langle a_i, \xi \rangle = s_i/\beta_i$; we absorb the bounded factor $1/\beta_i$ into the unit by rescaling s_i, keeping $|s_i| \asymp$ dist to the i-th surface). Write $\xi = (y, z)$ with $y \in V$ the stratum coordinates and $z = \sum_i s_i f_i \in W$ the pinned coordinates.

For the cap-volume bounds of Theorem 1.2 we work at fixed facet-distance θ, where by Lemma 2.2 every slope is bounded by the *fixed* constant $M_0 := 1/(\beta_* \theta)$ and every cosine $\beta_\bullet \geq \beta_* \theta$. Pin each constraint at scale h: define the *curved witness box* $\widehat{W}$ in coordinates (x_1, y, z) by

$$|y| \leq \eta \sqrt{h}, \qquad -\eta \, h \leq s_i \leq 0 \ \ (i = 1, \ldots, k), \qquad -h \leq x_1 \leq -\tfrac{h}{2}, \tag{7}$$

for a small structure constant $\eta > 0$ (we will need $\eta \leq \beta_* \theta$ and η small in absolute constants); $s_i \leq 0$ is the inward side of the i-th constraint. The pinned windows have length $\eta h \asymp h$ — this is the range that produces the factor h per constraint — so $|z| \leq F_0 \sum_i |s_i| \leq k F_0 \eta h \leq c\, h$, while the stratum coordinates carry $|y| \leq \eta \sqrt{h}$. Thus $|z| = O(h)$ but $|y| = O(\sqrt{h})$, and $|\xi|^2 = |y|^2 + |z|^2 \leq \eta^2 h + c^2 h^2 \leq c_1 \eta^2 h$ for h small.

$\widehat{W}$ **lies in the cap.** We verify $x_1 \le \Phi(\xi) = \min_\bullet \psi_\bullet(\xi)$ on $\widehat{W}$. The curvature term of each $\psi_\bullet$ is $\frac{1}{2}|\xi^\top Q_\bullet \xi| \le \frac{1}{2} C_* c_1 \eta^2 h \le \frac{h}{8}$ (η small), and the cubic remainder is $O((\eta^2 h)^{3/2}) \le \frac{h}{8}$ (h small). For the linear term $-\langle a_\bullet, \xi\rangle = -\langle a_\bullet|_V, y\rangle - \langle a_\bullet|_W, z\rangle$:

- *Constraint* $\bullet = i$. By the dual frame (6), $\langle a_i, \xi\rangle = -s_i/\beta_i$ has stratum part zero ($a_i|_V$ contributes through y only at quadratic order, absorbed above, since $P_u n_i \perp V$) and pinned part $-s_i/\beta_i \ge 0$ for $s_i \le 0$. Hence $-\langle a_i, \xi\rangle = s_i/\beta_i \le 0$: the pinning term only *helps* $x_1 \le \psi_i$. Its size is $|s_i|/\beta_i \le \eta h/(\beta_* \theta) \le h$ by $\eta \le \beta_* \theta$, well within budget.
- *Body* $\bullet = K$. Since the contact point is the deepest corner, 0 maximizes τ over K along the whole stratum, so $a_K|_V = 0$. The pinned part has slope $\le M_0$, so $|\langle a_K|_W, z\rangle| \le M_0|z| \le M_0 c\, h \le \frac{h}{8}$ (η small, since $|z| \le c\eta h$ carries the smallness).

Collecting, $\psi_\bullet(\xi) \ge -\frac{h}{8} - \frac{h}{8} - \frac{h}{8} \ge -\frac{h}{2} \ge x_1$ on $\widehat{W}$, so $\widehat{W} \subseteq C(h)$. The essential point is that the pinned coordinates sit at scale $h \ll \sqrt{h}$, so they contribute nothing at the curvature scale; only the stratum coordinates y reach scale $\sqrt{h}$, and there the body is critical ($a_K|_V = 0$) and contributes only its quadratic drop, itself $\le \frac{h}{8}$.

The pinning is genuinely linear. This is the heart of the curved estimate. Along the i-th constraint direction the floor is $\psi_i = s_i/\beta_i - \frac{1}{2}\xi^\top Q_i \xi + O(|\xi|^3)$. The *linear* coefficient of s_i is $1/\beta_i \ge 1$, bounded below, so the feasibility $\psi_i \ge -h$ permits $|s_i| \le \beta_i h + O(h^{3/2}) \asymp h$ (the window length used in (7)); the *quadratic* term $\frac{1}{2} Q_i s_i^2 = O(C_g h^2) \ll h$ cannot change this. This is the exact analogue of the linear constraint $x_i \le 0$ of [6] pinning $w_i \in [-h/(8ku_i), 0]$, with $u_i = \beta_i$; curvature perturbs the window only by a factor $1 + O(h)$. The mechanism that could change the exponent — the slope m_i growing to scale $h^{-1/2}$, so the linear coefficient $1/\beta_i$ itself reaches $h^{-1/2}$ and the window contracts toward $\sqrt{h}$ — does not occur at fixed θ, where $\beta_i \ge \beta_* \theta$ is bounded below; it is the subject of Section 4.

Volume. The k pinned coordinates s_i range over intervals of length ηh; the dual frame $\{f_i\}$ has bounded Gram, so the z-Jacobian is a structure constant; the j stratum coordinates y range over a ball of radius $\eta\sqrt{h}$; and the height fibre has length $\frac{h}{2}$. Hence

$$\mu(C(h)) \;\ge\; \frac{\mathrm{vol}(\widehat{W})}{\mathrm{vol}(B^n)} \;\ge\; c_2 \cdot \tfrac{h}{2} \cdot \prod_{i=1}^{k} (\eta h) \cdot (\eta\sqrt{h})^j \;=\; c_3\, h^{\,1+k+j/2} \;=\; c_3\, h^{\frac{2n-j}{2}}, \tag{8}$$

since $1 + k + j/2 = 1 + k + (n - 1 - k)/2 = (n + 1 + k)/2 = (2n - j)/2$. Here $c_3 \asymp \eta^{\,k+j}$ times structure constants; with the admissible choice $\eta = \beta_* \theta$ this gives $b_1(\theta) \asymp \theta^{\,k+j}$, the stated θ-dependence of the lower constant (positive for each fixed $\theta > 0$, degrading as $\theta \to 0$ — consistent with the cap growing as u nears $\partial\mathcal{U}_p$). This proves the lower bound of Theorem 1.2.

2.3 The upper bound

For the upper bound, by (3) the cap is contained in $\{-h \le x_1 \le \Phi(\xi)\} \cap \{\Phi(\xi) \ge -h\}$. Fix any single steep constraint, say K (steep for all admissible h). Its graph gives $x_1 \le \psi_K(\xi) = -\langle a_K, \xi\rangle - \frac{1}{2}\xi^\top Q_K \xi + O(|\xi|^3)$, and the height fibre has length at most $\psi_K(\xi) + h$, so

$$\mu(C(h)) \;\le\; \frac{1}{\mathrm{vol}(B^n)} \int_{\{\Phi \ge -h\}} \big(\Phi(\xi) + h\big)_+ \, d\xi \;\le\; \frac{1}{\mathrm{vol}(B^n)} \int_{\{\psi_K \ge -h\}} \big(\psi_K(\xi) + h\big)\, d\xi.$$

We bound the integration domain $\{\psi_K \geq -h\} \cap \bigcap_i \{\psi_i \geq -h\}$. The body constraint $\psi_K \geq -h$, by strict convexity $Q_K|_V \succeq c'_K I_V$ (Lemma 2.1) and $a_K|_V = 0$, forces $\frac{1}{2}c'_K|y|^2 \leq h + o(h)$, i.e. $|y| \leq C\sqrt{h}$ in the stratum directions. Each constraint $\psi_i \geq -h$, whose linear coefficient $1/\beta_i \geq 1$ in the pinned coordinate s_i is bounded below, forces $|s_i| \leq \beta_i h + O(h^{3/2}) \leq 2h$ (using $\beta_i \leq 1$) in the pinned directions. The integrand $\psi_K(\xi) + h$ is $\leq 2h$. Hence, with the bounded z-Jacobian of the dual frame,

$$\mu(C(h)) \;\leq\; C \cdot \underbrace{(C\sqrt{h})^{j}}_{y\text{-ball}} \cdot \underbrace{\prod_{i=1}^{k}(2h)}_{s_i\text{-intervals}} \cdot \underbrace{(2h)}_{\text{fibre}} \;=\; C'\, h^{j/2} \cdot h^{k} \cdot h \;=\; b_2\, h^{\frac{2n-j}{2}},$$

the exponent being $j/2 + k + 1 = (2n-j)/2$. Here the pinned length $|s_i| \leq 2h$ is curvature- and θ-independent (it uses only $\beta_i \leq 1$ and the lower bound $1/\beta_i \geq 1$ on the linear coefficient), so b_2 depends only on the structure constants — in particular the upper bound is uniform over the whole normal cone, not just at fixed facet-distance. This proves Theorem 1.2. □

Remark 2.3 (Why the constant no longer factors)**.** *In the simplicial model the cap constant factored as a product of a stratum-curvature factor (from Q_K) and one factor per truncation depending only on its angle, because the constraints were mutually orthogonal and the floor split as a sum of decoupled coordinates. With curved g_i the cross blocks of the floor $\sum_\bullet \frac{1}{2}\xi^\top Q_\bullet \xi$ couple the stratum directions y to the pinned directions z, and the dual frame (6) (which inverts the joint Gram of all active normals) mixes the truncation curvatures Q_i with the stratum curvature Q_K. Completing the square — shifting $y \mapsto y - Lz$ to remove the y–z cross block, with L bounded because $Q_K|_V \succeq c'_K I_V$ is invertible — moves y by $O(z) = O(h) \ll \sqrt{h}$ and changes the z-part at $O(h^2) \ll h$, so the product volume count survives with a constant depending on the joint form $Q_K|_V + \sum_i Q_i|_V$ rather than on Q_K and the Q_i separately. The exponent is untouched precisely because the shift is lower order at the cap scale.*

3 The slicing induction for curved slices

The cap-volume proposition of Section 2 handles the *stratum-direction* extremes ($u \in \mathcal{U}_p$). As in [6], the general direction is reduced to the stratum case by an induction that slices off one constraint at a time. In the simplicial model the slice of a half-space by a coordinate hyperplane is again a half-space, so the slice stays in class trivially. Here the slice is a curved hypersurface and we must verify it remains an admissible $(n-1, k-1)$ curved configuration with curvatures bounded for the fixed configuration.

3.1 The slice is in class

Fix a constraint, say g_{i_0}, and a level $s \in \mathbb{R}$ near 0. Let

$$\Sigma_s := \{x :\; g_{i_0}(x) = s\}, \qquad G_s^{(i_0)} := G \cap \Sigma_s = K \cap \Sigma_s \cap \bigcap_{i \neq i_0} \{g_i \leq 0\}.$$

Σ_s is a C^2 strictly convex hypersurface (principal curvatures in $[c_g/\Gamma_0,\, C_g/\gamma_0]$ by (1) and the formula for the curvature of a level set, $\kappa = (\nu^\top \operatorname{Hess} g_{i_0}\, \nu)/|\nabla g_{i_0}|$ for tangent ν). We regard $G_s^{(i_0)}$ as a subset of the $(n-1)$-manifold Σ_s, equipped with its induced Riemannian metric, and must produce on it the data of Definition 1.1 in dimension $n-1$ with $k-1$ constraints.

Lemma 3.1 (Slice is an admissible curved configuration)**.** *There are slice constants $c'_K \le C'_K$, $c'_g \le C'_g$, $\gamma' > 0$, $r'_0 > 0$ and $\delta_0 > 0$, depending on the fixed original configuration and on n, k, such that for every $|s| \le \delta_0$ the sliced configuration $G_s^{(i_0)} \subset \Sigma_s$ is admissible with structure constants $(c'_K, C'_K, c'_g, C'_g, \gamma', r'_0)$, with stratum $E_s^{(i_0)} := E \cap \Sigma_s$ of dimension $j - 1 = (n-1) - 1 - (k-1)$.*

Proof. We verify the three ingredients of Definition 1.1 for the body $K \cap \Sigma_s$ inside the manifold Σ_s and the constraints $\{g_i|_{\Sigma_s} \le 0\}$, $i \ne i_0$.

(a) The sliced body is strictly convex with bounded curvature. The body for the lower-dimensional section lemma is the convex region cut from the convex surface Σ_s by ∂K; its boundary in Σ_s is the transversal intersection $\partial K \cap \Sigma_s$ of the two strictly convex hypersurfaces ∂K and Σ_s. Its normal curvature in Σ_s is given by the same intersection-curvature formula (9) below, with II_{g_i} replaced by $\mathrm{II}_K \succeq c_K I$ and $\omega = \angle(\nu_K, \nu_{\Sigma_s})$ the dihedral angle, bounded away from $0, \pi$ by transversality. Since Σ_s is a truncation surface of K with the body on the inward side, the two surfaces are mutually convex ($\cos\omega \le 0$ generically), so the numerator is $\succeq c_K |X|^2$ and the sliced body has curvature in a band $[c'_K, C'_K]$ fixed by the original fixed-configuration data. This yields c'_K, C'_K.

(b) The restricted constraints are strictly convex hypersurfaces of Σ_s with bounded curvature. This is the delicate ingredient, and we state the mechanism, the regime where it is clean, and the regime where its *uniformity* is part of the open problem of §4. What we need is not that the function $\tilde{g}_i := g_i|_{\Sigma_s}$ be convex, but that the truncating *surface* $\Sigma'_i := \{g_i = 0\} \cap \Sigma_s$ be a strictly convex C^2 hypersurface of Σ_s, curving toward the kept region, with principal curvatures (intrinsic to Σ_s) bounded in a band $[c'_g, C'_g]$. The surface Σ'_i is the transversal intersection of the two convex hypersurfaces $\{g_i = 0\}$ and Σ_s. Its normal curvature in Σ_s, in a tangent direction $X \in T\Sigma'_i$, is given by the intersection-curvature formula

$$\kappa_{\Sigma'_i, \Sigma_s}(X) = \frac{\mathrm{II}_{g_i}(X,X) - \cos\omega\, \mathrm{II}_{\Sigma_s}(X,X)}{\sin\omega}, \qquad \omega := \angle\big(n_i, \nu_{\Sigma_s}\big), \tag{9}$$

There is only one power of $\sin\omega$: the unit conormal of the intersection inside Σ_s is $(n_i - \cos\omega\, \nu_{\Sigma_s})/\sin\omega$. where $\mathrm{II}_{g_i}, \mathrm{II}_{\Sigma_s} \succeq 0$ are the (convex) second fundamental forms of the two surfaces and ω is their dihedral angle, bounded away from 0 and π by the transversality constant γ (so $\sin\omega \ge \gamma'$, $|\cos\omega| \le 1 - \gamma''$). The numerator is $\succeq c_g\, |X|^2 - (1-\gamma'') C_g\, |X|^2$; this is the term that can fail to be positive if the truncation curvature c_g is small relative to C_g at an oblique angle.

Clean regime. If the two surfaces are *mutually convex* at the contact ($\cos\omega \le 0$, normals splaying apart, the generic transversal configuration for two truncations bounding a common region), then the numerator is $\succeq c_g |X|^2$ and $\kappa_{\Sigma'_i, \Sigma_s} \in [c_g, (C_g + C_g)/\gamma']$: the sliced surface is strictly convex with curvature bounded by structure constants, giving c'_g, C'_g. For our standing example (a ball truncated by spheres whose centers lie outside, so the truncation normals splay), and more generally whenever the truncations bound G from genuinely distinct directions, this is the operative case and the slice is unconditionally in class.

Oblique regime (uniformity caveat). When $\cos\omega > 0$ and $c_g < (1 - \gamma'') C_g$, (9) does not by itself certify positivity, and the strict convexity of Σ'_i uses the global convexity of $\{g_i \le 0\} \cap \Sigma_s$ rather than the pointwise curvature: the kept region is still convex (an intersection of convex sets restricted to the convex surface Σ_s), so Σ'_i is convex toward it, but its curvature lower bound c'_g now depends on the angle ω and degrades as $\omega \to 0$. For a *fixed* configuration ω is bounded away from 0 along the compact stratum, so $c'_g > 0$ is a genuine constant and the

induction proceeds; a bound on c'_g uniform over all families with the given structure constants, as ω ranges, is exactly Problem 4.12. The gradient bounds γ'_0, Γ'_0 are the slab-projections of ∇g_i, bounded by structure constants in this clean regime and by the fixed-configuration data in the general fixed-family regime.

(c) Transversality survives. The k outward normals $\{\nu_K, n_i : i \neq i_0\}$ restricted to $T\Sigma_s$ are the slab projections of the original normals onto the tangent space of Σ_s. Their Gram matrix is the original Gram matrix M with the row and column i_0 removed and the rest projected orthogonally to n_{i_0}; this is the Schur complement of M with respect to its i_0-th entry, which is $\succeq \gamma I$ because $M \succeq \gamma I$ and the Schur complement of a matrix $\succeq \gamma I$ along a unit direction is again $\succeq \gamma I$ on the orthogonal complement. Hence $\gamma' = \gamma$ works. Finally $G_s^{(i_0)}$ contains a ball of radius $r'_0 \geq c\, r_0$ in Σ_s for $|s| \leq \delta_0$, because the inner ball $B(c^-, r_0) \subseteq G$ of the original configuration meets the slab of slices in a cap of inradius $\geq c r_0$ inside Σ_s. □

3.2 The induction and the slab bound

With the slice in class, the induction of [6, §4, Case 2] runs verbatim with the affine slice replaced by the curved slice Σ_s and the flat radius $\sqrt{1-s^2}$ replaced by the bounded geometry of Lemma 3.1.

Proposition 3.2 (Cap mass, all directions)**.** *For each fixed configuration satisfying Definition 1.1 there are $b > 0$ and $h_0 > 0$, depending on that configuration, such that for every unit u with $\tau_{\max}(u)$ attained on E and every $0 < h \leq h_0$,*

$$\Lambda_u(h) := \mu\big(\{x \in G : \ \langle x, u\rangle \geq \tau_{\max}(u) - h\}\big) \ \geq \ b\, h^{\frac{2n-j}{2}}.$$

Proof sketch. Induction on k. The base case $k = 0$ (G a strictly convex body) is the rolling-ball cap bound of [5]: a cap of depth h on a body with principal curvatures in $[c_K, C_K]$ has the section identity $r(h)^2 = (\rho - g + h)(\rho + g - h) \geq h\rho$ with $\rho = 1/C_K$, hence mass $\geq c\, h^{(n+1)/2} = c\, h^{(2n-j)/2}$ at $j = n - 1$.

For the inductive step fix u with extreme on E. If $u \in \mathcal{U}_p$ at facet-distance $\geq \theta_0$ for a fixed threshold $\theta_0 > 0$ chosen from the fixed configuration data, Theorem 1.2 gives the bound directly. Otherwise u is within θ_0 of a facet of the normal cone. Write $u = \sum_\bullet \lambda_\bullet \nu_\bullet$ with $\lambda^\top M \lambda = 1$, $M \succeq \gamma I$; then $\max_\bullet \lambda_\bullet \geq (k+1)^{-1}\gamma^{1/2}$, so *some* active constraint g_{i_0} has a multiplier bounded below, hence (by $\beta_{i_0} = (M\lambda)_{i_0} \geq \gamma \lambda_{i_0}$) angle bounded away from $\pi/2$: $\beta_{i_0} \geq c > 0$ a structure constant. This strongly-active constraint is the one to slice (exactly as [6] slices the coordinate i_0 with u_{i_0} bounded below in its Case 2). For s in an interval of length $\asymp h$ below the extreme level, the slice $G_s^{(i_0)}$ is an admissible $(n-1, k-1)$ curved configuration (Lemma 3.1) and the restricted direction $u' := P_{\Sigma_s} u / |P_{\Sigma_s} u|$ has its extreme on the sliced stratum $E_s^{(i_0)}$, at depth $h_* \asymp h$. By the induction hypothesis applied in dimension $n-1$ with $k-1$ constraints (stratum dimension $(n-1) - 1 - (k-1) = j - 1$),

$$\Lambda_{u'}^{(n-1,k-1)}(h_*) \ \geq \ b'\, h^{\frac{2(n-1)-(j-1)}{2}} = b'\, h^{\frac{2n-j-1}{2}}.$$

Integrating over the slicing interval ($\asymp h$) by the curved coarea formula (the Jacobian $|\nabla g_{i_0}|$ is bounded above and below by (1), so the coarea factor is a structure constant) gives

$$\Lambda_u(h) \ \geq \ c \cdot h \cdot b'\, h^{\frac{2n-j-1}{2}} \ = \ c'\, h^{\frac{2n-j}{2}}.$$

The exponent telescopes exactly as in the flat case: one slice supplies a factor h (the slicing interval), and the induction supplies $h^{(2n-j-1)/2}$, so the product is $h^{1+(2n-j-1)/2} = h^{(2n-j)/2}$, with the codimension count $j \to j-1$ matching $(n,k) \to (n-1,k-1)$. □

Proof of Theorem 1.3. This is the anchor–unimodality conversion of [5, 6], which uses only: (i) the cap-mass bound of Proposition 3.2 at both u-extremes; (ii) the unimodality of the level-section area A_u, which holds by Brunn–Minkowski because G is convex; and (iii) the inner ball $B(c^-, r_0) \subseteq G$ providing a bulk anchor of value $a_1(n,k) > 0$ at a depth $h^\dagger \geq c(k) > 0$. Define $A_u(h) := \mathrm{vol}_{n-1}(G \cap \{\langle x, u\rangle = \tau_{\max} - h\})$. Below the mode of A_u, $A_u(h) \geq \Lambda_u(h)/h \geq b\, h^{(2n-j-1)/2}$ by Proposition 3.2; above the mode, unimodality gives $A_u(h) \geq A_u(h^\dagger) \geq a_1$. Anchoring at depth $L/4$ from each of the u- and $(-u)$-extremes and integrating over the middle of the τ-range (whose complement has length $\leq L/2$, while the range has length $\geq 2r_0 \geq 2L$) yields $\mathrm{vol}(G \cap \{|\langle y - x, u\rangle| \leq L\}) \geq a_1 L^{(2n-j)/2}$. □

Proof of Corollary 1.4. Upper bound. Insert Theorem 1.3 into the proof skeleton of [3, 5]: the population dichotomy holds with the inner-ball radius r_0; the VC relative deviation bounds apply to slabs cut by k fixed C^2 convex constraints, whose graphs $\{g_i \leq 0\}$ together with two parallel slab walls form a family of sets of finite VC dimension (the class $\{g_i \leq t\}$ is a translate of a fixed graph, hence a VC class, and finite intersections multiply the dimension); thus a sample point in S_w yields $\mu(S_{\kappa w}) \geq 32 u_N$ for all $w \geq \ell_N := C(u_N)^{2/(2n-j)} \asymp (\log N/N)^{2/(2n-j)}$, and the UDC holds with probability $1 - 1/N$.

Lower bound. The second-moment scheme of [5, 6] applies with the curved stratum caps of Theorem 1.2 in place of the flat ones. The cap directions are the stratum directions $u_p \in \mathcal{U}_p$ at a fixed facet-distance θ_0 (any fixed interior direction of the normal cone), so Theorem 1.2 gives two-sided cap masses $\asymp h^{(2n-j)/2}$ with θ_0-dependent but N-independent constants. The footprint of a depth-H cap is $\asymp \sqrt{H}$ (the y-ball radius), so a maximal $2c\sqrt{H_N}$-separated family on the j-manifold E has $M \geq c H_N^{-j/2} \geq c'(N/\tau_N)^{j/(2n-j)}$ members with pairwise disjoint outer caps. The first/second moment computation is identical to [6]: with $\tau_N = \varepsilon \log N$ chosen so that the per-cap gap probability is $N^{-2q\varepsilon}$ with $q = (4\kappa^2)^{(2n-j)/2} b_2/b_1$, one gets $\mathbb{E}X \to \infty$ and $\mathbb{P}(X = 0) = o(1)$ by Chebyshev, forcing $\ell \geq c(\log N/N)^{2/(2n-j)}$. □

4 What uniformity is and is not obtained

We now state precisely the scope of Corollary 1.4 and the residual open problem.

4.1 Obtained: fixed family, uniform over the stratum and compact subcones

For a *fixed* admissible configuration (a fixed body K and fixed C^2 strictly convex $g_1, \ldots, g_k$), Corollary 1.4 gives the exact law $\Theta((\log N/N)^{2/(2n-j)})$. The constants in Theorems 1.2–1.3 are uniform:

- *over the stratum* $p \in E$: the structure constants $(c_K, C_K, c_g, C_g, \gamma_0, \Gamma_0, \gamma, r_0)$ and the C^2 Taylor remainders are uniform on the compact manifold E;
- *over directions* u in any fixed compact subcone of the normal cone (facet-distance $\geq \theta_0$): the cap exponent is exactly $(2n-j)/2$, with the constant determined by the fixed configuration and θ_0, for all depths $h \leq h_1 \theta_0^2$. The upper bound constant b_2 is uniform

over the cone, while the lower bound constant $b_1(\theta_0)$ degrades as the subcone approaches the boundary; this is the loss coming from the pinned lengths $s_i \le Ch/(\beta_*\theta_0)$.

This is the regime needed for both directions of the law: the slab-volume bound (Theorem 1.3) is a statement for *all* u, obtained by the slicing induction which always slices the most binding constraint and so always operates at definite angle; and the lower-bound caps may be taken at a single fixed interior direction $\theta_0 = \frac{1}{2}\operatorname{diam}\mathcal{U}_p$, say.

4.2 The exponent is curvature-robust but direction-sensitive

The exponent $(2n-j)/2$ does not depend on the magnitudes of the curvatures: for a fixed active set and fixed interior direction the slopes m_i are fixed constants, so $h \to 0$ keeps every constraint steep ($m_i \le C_*^{-1/2}h^{-1/2}$ automatically once h is small), and each pinned coordinate contributes a clean factor h regardless of Q_i. High truncation curvature only inflates the constant (through $\sqrt{h/c_g}$ in the shallow regime and the joint form in Remark 2.3); it never changes the exponent.

The exponent *does* change as u approaches the boundary of the normal cone $\mathcal{U}_p$, and this is correct rather than a defect. Two equivalent descriptions of the boundary: a multiplier vanishes ($\lambda_{i_0} \to 0$, the i_0-th constraint becomes first-order inactive at the maximizer), or equivalently the slab tilts toward tangency with surface i_0 ($\beta_{i_0} = \langle n_{i_0}, u\rangle \to 0$, slope $m_{i_0} \to \infty$), so that p ceases to be the strict τ-maximum and the extreme spreads onto a *larger* stratum E' of dimension $j+1$. There the cap is governed by the $(2n-j')/2$ law with $j' = j+1$: a *larger* cap. The natural single formula across the closed cone is the *trichotomy interpolant*: with the per-surface pinned length

$$\lambda_i(h) := \min\Big(\frac{h}{m_i},\ \sqrt{h/c_g}\Big), \qquad m_i = \tan\angle(n_i, u),$$

one expects

$$\mu(C(h)) \asymp h^{1+j/2}\prod_{i=1}^{k}\lambda_i(h), \tag{10}$$

which interpolates: when all m_i are $O(1)$, $\lambda_i(h) \asymp h$ and (10) is $\asymp h^{1+j/2}\cdot h^k = h^{(2n-j)/2}$; when m of the slopes exceed $h^{-1/2}$ (those $\beta_i \lesssim \sqrt{h}$), the curvature cap gives $\lambda_i(h) \asymp \sqrt{h}$ and the exponent rises to $(2n-(j+m))/2$, exactly matching the larger stratum that has become active. The crossover is at $m_i \asymp h^{-1/2}$, i.e. $\beta_i \asymp \sqrt{h}$, the same $\sqrt{h}$ angular threshold as the simplicial trichotomy $\mathcal{A}/\mathcal{Z}$ of [6] (there $\lambda_i = \min(h/(8ku_i), \sqrt{h}/8)$ with $u_i = \beta_i$); curvature does not move the threshold, since the shallow length $\sqrt{h/c_g}$ and the steep length h/m_i are sandwiched between the flat-case values with constants from $[c_g, C_g]$. We obtain (10) as a valid *upper* bound for all u (the per-surface lengths $\lambda_i(h)$ each cap the i-th coordinate independently); the matching lower bound is Theorem 1.2 away from $\partial\mathcal{U}_p$, and is open in a shrinking neighborhood of the multiply-tangent directions, where the pinned block can degenerate jointly even though each λ_i is controlled.

4.3 Open: uniformity over curvature-bounded families

The hypotheses we cannot remove — and which constitute the residual open problem — concern uniformity *across a family* of configurations rather than for a single fixed one. Two distinct gaps:

Problem 4.1 (Uniform constant near multiply-tangent directions)**.** *Let* $\mathcal{F}(c_K, C_K, c_g, C_g, \gamma_0, \Gamma_0, \gamma, r_0)$ *be the class of* all *admissible configurations with the given structure constants. Is the slab-volume*

bound of Theorem 1.3 uniform over $\mathcal{F}$ — i.e. are a_1, L_0 functions of the structure constants alone, with no dependence on the individual configuration? Equivalently, in (10), is the lower *bound uniform as u approaches a codimension-≥ 2 face of the normal cone, where two or more $\beta_i \to 0$ simultaneously?*

The obstruction is genuine. When a single $\beta_{i_0} \to 0$, the slicing induction handles it (slice on g_{i_0}). When two angles $\beta_{i_0}, \beta_{i_1} \to 0$ together, the slab projections $P_u n_{i_0}, P_u n_{i_1}$ of the two near-tangent normals can themselves become nearly parallel, so the dual frame (6) blows up, $|f_i| \to \infty$, and the joint Gram of the pinned block degrades. The trichotomy bound (10) remains a valid *upper* bound there (it only uses the diagonal β_i), but the matching *lower* bound requires the joint Gram of $\{P_u n_i : \beta_i \text{ small}\}$ to stay bounded below, which is an extra non-degeneracy hypothesis not implied by the per-surface structure constants. For a fixed configuration this is automatic (the finite set of multiply-tangent directions has its own positive Gram lower bound, absorbed into the configuration's constants); uniformly over $\mathcal{F}$ it is open, because the Gram lower bound can be made to deteriorate within the class while keeping the structure constants fixed.

There is, however, a clean conditional positive statement. It isolates exactly the extra geometry that the proof needs beyond the per-surface structure constants.

Definition 4.2 (Resolved multi-tangent geometry)**.** *Fix structure constants and numbers $\Gamma, c_{\mathrm{sl}} > 0$. A subfamily $\mathcal{F}_{\Gamma,c_{\mathrm{sl}}} \subseteq \mathcal{F}(c_K, C_K, c_g, C_g, \gamma_0, \Gamma_0, \gamma, r_0)$ is called* $(\Gamma, c_{\mathrm{sl}})$-resolved *if the following two properties hold for every configuration in the subfamily and for every iterated one-constraint slice that can occur in the induction of Lemma 3.1.*

(R1) Every sliced truncating surface remains strictly convex with lower principal curvature at least c_{sl} and with upper curvature and gradient bounds controlled by the original structure constants and c_{sl}^{-1}.

(R2) Let p be a point of the active stratum, let u lie in the closed normal cone at p, and let P_u denote orthogonal projection onto $u^\perp$. For every subset I of active curved truncating normals for which $P_u n_i(p) \neq 0$, the normalized projected vectors

$$e_i(u,p) := \frac{P_u n_i(p)}{|P_u n_i(p)|}, \qquad i \in I,$$

have Gram matrix bounded below by ΓI.

Condition (R1) is automatic in the clean slicing regime of Lemma 3.1; condition (R2) is automatic for a fixed configuration but is not forced uniformly by the structure constants alone.

Proposition 4.3 (Uniform curved law under resolved multi-tangent geometry)**.** *On every $(\Gamma, c_{\mathrm{sl}})$-resolved subfamily $\mathcal{F}_{\Gamma,c_{\mathrm{sl}}}$, the slab bound of Theorem 1.3 is uniform: there are $a_*, L_* > 0$, depending only on the structure constants, n, k, Γ, and c_{sl}, such that every configuration in $\mathcal{F}_{\Gamma,c_{\mathrm{sl}}}$ satisfies*

$$\mathrm{vol}\Big(G \cap \{y : \ |\langle y - x, u\rangle| \leq L\}\Big) \geq a_* \, L^{\frac{2n-j}{2}}$$

for every $x \in G$, every unit u, and every $0 < L \leq L_*$. Consequently the interpolation law of Corollary 1.4 holds uniformly over $\mathcal{F}_{\Gamma,c_{\mathrm{sl}}}$, with UDC constants depending only on the displayed parameters and not on the individual curved configuration.

Proof. Repeat the proof of Proposition 3.2 and Theorem 1.3, tracking constants. Away from a face of the normal cone, the lower witness box in Theorem 1.2 only uses bounded Taylor remainders, the curvature bands, and the bounded inverse of the Gram matrix of the projected active normals. Condition (R2) supplies that inverse bound uniformly, so the dual frame (6) has $|f_i| \le C(\Gamma, k)$ throughout the resolved subfamily.

Near a face, the induction slices on a strongly active constraint exactly as in the proof of Proposition 3.2. Condition (R1) says that the sliced configuration remains in the same quantitative class, while the Schur-complement transversality argument in Lemma 3.1 gives the same lower Gram bound for the remaining unsliced normals. Condition (R2), imposed on all iterated slices, prevents the only remaining loss: simultaneous projected-normal degeneration in the pinned block. Hence the inductive constants in the cap lower bound are functions only of n, k, the structure constants, Γ, and c_{sl}. The coarea interval has length $\asymp h$ with constants controlled by the gradient bounds, so the exponent telescopes as before and yields a uniform version of Proposition 3.2.

The conversion from uniform cap mass to the uniform slab estimate is the same anchor–unimodality argument used in Theorem 1.3; the inner-ball radius is part of the structure constants. Finally the empirical UDC proof skeleton of the companion papers uses only this slab constant, the fixed VC dimension of the family, and the density bounds, so the high-probability interpolation law is uniform on the resolved subfamily. □

Corollary 4.4 (Finite curved atlases are uniform)**.** *Let $\mathcal{F}_0 = \{(K^a, g_1^a, \dots, g_k^a) : 1 \le a \le M\}$ be a finite collection of admissible curved configurations with the same n, k. After enlarging the structure constants to dominate the finitely many members, there are $a_0, L_0 > 0$, depending on the finite atlas $\mathcal{F}_0$, such that every configuration in $\mathcal{F}_0$ satisfies*

$$\mathrm{vol}\Big(G^a \cap \{y : \ |\langle y - x, u\rangle| \le L\}\Big) \ge a_0 \, L^{\frac{2n-j}{2}}$$

for every a, every $x \in G^a$, every unit u, and every $0 < L \le L_0$. Consequently, for any common density bounds, the interpolation law of Corollary 1.4 holds uniformly over the whole finite atlas.

Proof. For each fixed member of the atlas, the discussion following Definition 4.2 gives constants $\Gamma_a, c_{\mathrm{sl},a} > 0$ for which that single configuration is $(\Gamma_a, c_{\mathrm{sl},a})$-resolved: compactness and fixed-configuration admissibility give positive Gram minima for the projected normals and positive curvature floors for the slices. Since the atlas is finite,

$$\Gamma_0 := \min_a \Gamma_a > 0, \qquad c_0 := \min_a c_{\mathrm{sl},a} > 0.$$

Thus $\mathcal{F}_0$ is (Γ_0, c_0)-resolved after using common structure constants for all members. Proposition 4.3 gives the displayed slab estimate and the uniform interpolation law. □

Corollary 4.5 (Compact resolved curved atlases are uniform)**.** *Let T be a compact parameter space, and let*

$$\mathcal{F}_T = \{(K^\tau, g_1^\tau, \dots, g_k^\tau) : \tau \in T\}$$

be a family of admissible curved configurations with the same n, k and common structure constants. Suppose there are continuous functions

$$\Gamma : T \to (0, \infty), \qquad c_{\mathrm{sl}} : T \to (0, \infty),$$

such that the configuration indexed by τ is $(\Gamma(\tau), c_{\mathrm{sl}}(\tau))$-resolved. Then there are $a_0, L_0 > 0$, depending on the compact atlas $\mathcal{F}_T$, such that every $\tau \in T$ satisfies

$$\mathrm{vol}\Big(G^\tau \cap \{y : \ |\langle y - x, u\rangle| \le L\}\Big) \ge a_0\, L^{\frac{2n-j}{2}}$$

for every $x \in G^\tau$, every unit u, and every $0 < L \le L_0$. Consequently, for any common density bounds, the interpolation law of Corollary 1.4 holds uniformly over the whole compact atlas.

Proof. By compactness and continuity,

$$\Gamma_0 := \min_{\tau \in T} \Gamma(\tau) > 0, \qquad c_0 := \min_{\tau \in T} c_{\mathrm{sl}}(\tau) > 0.$$

Thus the whole atlas is (Γ_0, c_0)-resolved. Apply Proposition 4.3. □

Corollary 4.6 (C^2-neighborhoods of resolved configurations)**.** *Let*

$$\{(K^\tau, g_1^\tau, \dots, g_k^\tau) : \tau \in T\}$$

be a parametrized family of admissible curved configurations with common structure constants, and suppose the data vary continuously in C^2 in the following local sense: after choosing finitely many graph charts for the active stratum and for every iterated one-constraint slice occurring in Lemma 3.1, the graph functions, their first derivatives, and their Hessians depend continuously on τ on compact chart domains. If the configuration at τ_0 is (Γ_0, c_0)-resolved, then there is a neighborhood U of τ_0 such that every configuration with $\tau \in U$ is $(\Gamma_0/2, c_0/2)$-resolved, after possibly restricting the slice-depth parameter in Lemma 3.1 by a factor depending only on the common structure constants and on the chosen charts.

Consequently, on U the slab bound of Theorem 1.3 and the interpolation law of Corollary 1.4 hold uniformly, with constants depending only on the common structure constants, n, k, Γ_0, and c_0.

Proof. For the fixed configuration τ_0, the quantities in (R1) and (R2) are positive minima of continuous functions over compact incidence spaces: the active stratum is compact, there are only finitely many orders in which the induction can slice the k constraints, the permitted slice parameters lie in a compact interval after the standard depth restriction, the unit normal directions and pinned subsets range over compact sets, and the list of active subsets is finite. By hypothesis these functions vary continuously in the chosen C^2 charts. Hence their minima are lower semicontinuous at τ_0. After shrinking the parameter neighborhood U, all sliced curvature floors stay above $c_0/2$, and all projected-normal Gram minima in (R2) stay above $\Gamma_0/2$. Upper curvature, gradient, and transversality bounds remain controlled because the family has common structure constants and the charts vary C^2-continuously. Thus the nearby configurations are $(\Gamma_0/2, c_0/2)$-resolved. Proposition 4.3 then gives the uniform slab and UDC conclusions. □

Corollary 4.7 (Compact C^2-continuous resolved atlases)**.** *Let T be compact, and let*

$$\mathcal{F}_T = \{(K^\tau, g_1^\tau, \dots, g_k^\tau) : \tau \in T\}$$

be a C^2-continuous atlas in the sense of Corollary 4.6, with common structure constants. Assume that each individual configuration in the atlas is resolved in the sense of Definition 4.2. Then there are $a_0, L_0 > 0$ such that every $\tau \in T$ satisfies

$$\mathrm{vol}\Big(G^\tau \cap \{y : \ |\langle y - x, u\rangle| \le L\}\Big) \ge a_0\, L^{\frac{2n-j}{2}}$$

for every $x \in G^\tau$, every unit u, and every $0 < L \le L_0$. Consequently, for any common density bounds, the interpolation law of Corollary 1.4 holds uniformly over the whole compact atlas.

Proof. For each $\tau \in T$, choose resolved constants (Γ_τ, c_τ) for that single configuration. By Corollary 4.6, there is a neighborhood U_τ on which the atlas is $(\Gamma_\tau/2, c_\tau/2)$-resolved. Compactness gives a finite subcover $U_{\tau_1}, \dots, U_{\tau_M}$. Set

$$\Gamma_* := \frac{1}{2} \min_{1 \le a \le M} \Gamma_{\tau_a} > 0, \qquad c_* := \frac{1}{2} \min_{1 \le a \le M} c_{\tau_a} > 0.$$

Then the whole atlas is (Γ_*, c_*)-resolved after using the common structure constants. Proposition 4.3 gives the uniform slab estimate and the uniform interpolation law. □

Corollary 4.8 (Exhaustive resolved curved atlases)**.** *Let $T = \bigcup_{q \ge 1} T_q$, where $T_q \subset T_{q+1}$ are compact parameter spaces, and let*

$$\mathcal{F}_T = \{(K^\tau, g_1^\tau, \dots, g_k^\tau) : \tau \in T\}$$

be a family of admissible curved configurations with the same n, k and common structure constants. Suppose there are functions

$$\Gamma : T \to (0, \infty), \qquad c_{\mathrm{sl}} : T \to (0, \infty),$$

continuous on each T_q, such that the configuration indexed by τ is $(\Gamma(\tau), c_{\mathrm{sl}}(\tau))$-resolved. Then, for each q, there are constants $a_q, L_q > 0$ such that

$$\mathrm{vol}\Big(G^\tau \cap \{y : \ |\langle y - x, u\rangle| \le L\}\Big) \ge a_q \, L^{\frac{2n-j}{2}}$$

for every $\tau \in T_q$, every $x \in G^\tau$, every unit u, and every $0 < L \le L_q$. Consequently, for any common density bounds, the interpolation law of Corollary 1.4 holds uniformly on each compact stage T_q.

If, in addition,

$$\inf_{\tau \in T} \Gamma(\tau) > 0, \qquad \inf_{\tau \in T} c_{\mathrm{sl}}(\tau) > 0,$$

then a_q, L_q may be chosen independently of q, and the interpolation law is uniform over the whole atlas $\mathcal{F}_T$.

Proof. For each fixed q, Corollary 4.5 applies to the compact subatlas over T_q, giving constants a_q, L_q. If the two displayed infima over T are positive, then the whole family lies in one resolved class with

$$\Gamma_0 = \inf_{\tau \in T} \Gamma(\tau), \qquad c_0 = \inf_{\tau \in T} c_{\mathrm{sl}}(\tau).$$

Proposition 4.3 gives one pair of constants for all $\tau \in T$. □

Corollary 4.9 (Uniform one-cut curved law under splaying)**.** *Fix structure constants as in Definition 1.1 and fix $\chi > 0$. Let $\mathcal{F}_{1,\chi}$ be the class of admissible curved configurations with $k = 1$ such that, for every $p \in E$,*

$$\langle \nu_K(p), n_1(p) \rangle \le -\chi.$$

Then the slab bound of Theorem 1.3 is uniform on $\mathcal{F}_{1,\chi}$: there are $a_\chi, L_\chi > 0$, depending only on n, the structure constants, and χ, such that every configuration in $\mathcal{F}_{1,\chi}$ satisfies

$$\mathrm{vol}\Big(G \cap \{y : \ |\langle y - x, u\rangle| \le L\}\Big) \ge a_\chi \, L^{\frac{n+2}{2}}$$

for every $x \in G$, every unit u, and every $0 < L \le L_\chi$. Consequently the interpolation law is uniform on $\mathcal{F}_{1,\chi}$:

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(n+2)}$$

with constants depending only on n, the density bounds, the structure constants, and χ.

Proof. For $k = 1$ there is no simultaneous multi-tangent block of curved truncating normals. Thus condition (R2) in Definition 4.2 is automatic with $\Gamma = 1$, since every relevant projected-normal family has at most one unit vector.

It remains to check that the one possible slice has uniform geometry. Slice by $g_1 = s$. After this slice there are no remaining truncating surfaces, so the only lower-dimensional body is $K \cap \Sigma_s \subset \Sigma_s$, where $\Sigma_s = \{g_1 = s\}$. The splaying condition and the C^2 bounds imply, after choosing $|s| \le \delta_\chi$ with δ_χ depending only on the displayed parameters, that the dihedral angle between ∂K and Σ_s still satisfies

$$\cos\omega \le -\chi/2, \qquad \sin\omega \ge c(\gamma,\chi) > 0.$$

The intersection-curvature formula (9), with II_{g_i} replaced by II_K, then gives

$$\kappa_{\partial K\cap\Sigma_s,\Sigma_s}(X) = \frac{\mathrm{II}_K(X,X) - \cos\omega\,\mathrm{II}_{\Sigma_s}(X,X)}{\sin\omega} \ge c_\chi |X|^2$$

and also an upper curvature bound $C_\chi|X|^2$, with constants depending only on the structure constants and χ. The sliced inradius and transversality bounds are inherited from the original inner-ball and Gram assumptions, as in Lemma 3.1. Hence the family $\mathcal{F}_{1,\chi}$ is $(1, c_\chi)$-resolved in the sense of Definition 4.2. Proposition 4.3 gives the uniform slab estimate and the uniform interpolation law. Since $j = n - 1 - k = n - 2$, the exponent is $2/(2n - j) = 2/(n + 2)$. □

Corollary 4.10 (Uniform one-cut curved law under curvature dominance)**.** *Fix structure constants as in Definition 1.1 and fix $\chi > 0$. Let $\mathcal{F}^{\mathrm{dom}}_{1,\chi}$ be the class of admissible curved configurations with $k = 1$ such that, for every $p \in E$,*

$$c_K - \max\{\langle \nu_K(p), n_1(p)\rangle, 0\}\,\frac{C_g}{\gamma_0} \ge \chi.$$

Then the slab bound of Theorem 1.3 is uniform on $\mathcal{F}^{\mathrm{dom}}_{1,\chi}$: there are $a_{\mathrm{dom}}, L_{\mathrm{dom}} > 0$, depending only on n, the structure constants, and χ, such that every configuration in $\mathcal{F}^{\mathrm{dom}}_{1,\chi}$ satisfies

$$\mathrm{vol}\Big(G \cap \{y :\ |\langle y - x, u\rangle| \le L\}\Big) \ge a_{\mathrm{dom}}\, L^{\frac{n+2}{2}}$$

for every $x \in G$, every unit u, and every $0 < L \le L_{\mathrm{dom}}$. Consequently the interpolation law is uniform on $\mathcal{F}^{\mathrm{dom}}_{1,\chi}$:

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(n+2)}.$$

In particular, if $c_K > C_g/\gamma_0$, then every one-cut admissible family with the given structure constants is covered, with $\chi = c_K - C_g/\gamma_0$.

Proof. As in Corollary 4.9, condition (R2) of Definition 4.2 is automatic for $k = 1$, because there is at most one projected truncating normal in any pinned block. We only need a uniform lower curvature bound for the single slice $K \cap \Sigma_s \subset \Sigma_s$, where $\Sigma_s = \{g_1 = s\}$.

The level-surface curvature bound in the proof of Lemma 3.1 gives $\mathrm{II}_{\Sigma_s} \preceq (C_g/\gamma_0)I$. The margin assumption and the C^2 bounds imply that, after restricting to $|s| \le \delta_\chi$ with δ_χ depending only on the displayed parameters,

$$c_K - \max\{\cos\omega, 0\}\,\frac{C_g}{\gamma_0} \ge \chi/2, \qquad \sin\omega \ge c(\gamma) > 0,$$

for the dihedral angle ω between ∂K and Σ_s. Therefore the intersection-curvature formula (9), again with II_{g_i} replaced by II_K, yields

$$\kappa_{\partial K \cap \Sigma_s, \Sigma_s}(X) = \frac{\mathrm{II}_K(X,X) - \cos\omega\, \mathrm{II}_{\Sigma_s}(X,X)}{\sin\omega} \ge c_{\mathrm{dom}}|X|^2,$$

with $c_{\mathrm{dom}} > 0$ depending only on the structure constants and χ. The corresponding upper curvature, inradius, and transversality bounds are controlled in the same way as in Lemma 3.1. Hence $\mathcal{F}^{\mathrm{dom}}_{1,\chi}$ is $(1, c_{\mathrm{dom}})$-resolved, and Proposition 4.3 gives the uniform slab estimate and the uniform interpolation law. The final assertion follows because $\max\{\langle \nu_K, n_1\rangle, 0\} \le 1$. □

Corollary 4.11 (Uniform one-cut law under nonobtuse contact)**.** *Fix structure constants as in Definition 1.1. Let $\mathcal{F}^{\mathrm{no}}_1$ be the class of admissible curved configurations with $k = 1$ such that, for every $p \in E$,*

$$\langle \nu_K(p), n_1(p)\rangle \le 0.$$

Then the slab bound of Theorem 1.3 is uniform on $\mathcal{F}^{\mathrm{no}}_1$: there are $a_{\mathrm{no}}, L_{\mathrm{no}} > 0$, depending only on n and the structure constants, such that every configuration in $\mathcal{F}^{\mathrm{no}}_1$ satisfies

$$\mathrm{vol}\Big(G \cap \{y : \ |\langle y - x, u\rangle| \le L\}\Big) \ge a_{\mathrm{no}}\, L^{\frac{n+2}{2}}$$

for every $x \in G$, every unit u, and every $0 < L \le L_{\mathrm{no}}$. Consequently the interpolation law is uniform on $\mathcal{F}^{\mathrm{no}}_1$:

$$\ell_N \asymp \Big(\frac{\log N}{N}\Big)^{2/(n+2)}.$$

Proof. For every configuration in $\mathcal{F}^{\mathrm{no}}_1$,

$$c_K - \max\{\langle \nu_K(p), n_1(p)\rangle, 0\}\frac{C_g}{\gamma_0} \ge c_K \qquad (p \in E).$$

Thus the class is contained in $\mathcal{F}^{\mathrm{dom}}_{1,c_K/2}$. Corollary 4.10 gives the uniform slab estimate and interpolation law. □

Problem 4.12 (Separation of curvatures)**.** *In the simplicial model the cap constant factored through the stratum curvature Q and the truncation angles separately. Quantify the failure for curved data: find the sharp dependence of b_1, b_2 on the joint second fundamental form $Q_K|_V + \sum_i Q_i|_V$ versus the separate forms, and determine whether a factorized lower bound $b_1 \ge c\,(\det Q_K|_V)^{-1/2} \prod_i \phi(\mathrm{angle}_i)$ holds with ϕ independent of the cross-curvature. Remark 2.3 shows the cross blocks are lower order at the cap scale, so a factorized bound holds* up to a multiplicative constant depending on the condition number of the joint form*; whether that constant can be removed — giving a clean product formula as in the flat case — is open.*

We summarize the honest status. *Fixed smooth strictly convex family, in general position, transversal along a j-stratum: the interpolation law $\Theta((\log N/N)^{2/(2n-j)})$ holds, with the same exponent and logarithm as the flat case, and constants determined by the configuration's curvature and transversality data* (Corollary 1.4). Away from multiply-tangent directions the estimates are controlled by the stated nondegeneracy constants; Corollary 4.9 gives a fully uniform one-curved-cut theorem under quantitative splaying, and Corollary 4.10 gives the same uniform law in the curvature-dominated oblique one-cut regime, with Corollary 4.11 recording the nonobtuse-contact specialization. The atlas corollaries, Corollaries 4.4 and 4.5, cover any finite menu of fixed curved geometries, and any compact atlas whose resolved constants vary continuously and stay positive; the local stability and finite-subcover form in Corollaries 4.6 and 4.7 show that resolved C^2-continuous compact atlases are uniform even when one checks the resolved geometry point by point. The remaining curvature-band-uniform slab/lower constants are open only where these one-cut slice-curvature criteria fail, where no compact C^2-resolved atlas controls the degeneration, or in the shrinking multi-tangent regime (Problems 4.1–4.12). As a fixed-family assertion, the exponent is settled for every first-order active stratum: it is $(2n-j)/2$ for that stratum, and the trichotomy bound (10) is the fixed-configuration interpolant across strata.

5 Concluding remarks

The curved case isolates exactly which features of the simplicial proof are essential and which are conveniences. The orthogonality $\langle w, \tilde{u}\rangle = 0$ of the orthogonal model and the biorthogonal frame $\langle f_i, a_{i'}\rangle = \delta_{ii'}$ of the general-position simplicial proof are conveniences: both are devices to decouple the k constraints in the level functional, and both are replaced here by a single completing-the-square in the slab coordinates, valid because the cross blocks of the joint second fundamental form are lower order ($O(h)$) at the cap scale ($\sqrt{h}$). The essential feature is the *linear pinning*: each transversal constraint contributes a full factor h through its first-order part, and this survives curvature because the strictly convex correction is quadratic, hence $O(h^2)$ in the pinned variable. The only place curvature could change the exponent is the near-tangent regime $\beta_i \lesssim \sqrt{h}$, and there it does not change the exponent of the relevant stratum — it relabels which stratum is active, with the trichotomy threshold $\sqrt{h}$ identical to the flat case.

Within the stability theory of [1, 2] the conclusion is the same as for flat truncations for each fixed smooth activation geometry: the finest resolution at which i.i.d. data certify the doubling condition is set by the most isolated extremal sample point of the lowest transversal stratum of the truncation geometry, with the exponent ladder $\frac{2}{n+1}, \dots, \frac{1}{n}$ governed by the stratum dimension and *not* by the curvature of the truncating surfaces. The remaining quantitative question — a curvature-band- and transversality-band-uniform lower/slab constant near the multiply-tangent directions — is the honest residue, and is the natural next target.

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Declaration on generative AI

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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